Graph Signal Processing and Interpolation for EEG Channel Reconstruction

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***Abstract*—This paper studies EEG channel reconstruction under missing-channel conditions using a multi-dataset Graph Signal Processing (GSP) framework. While conventional geometric baselines typically focus on spatial smoothing, recent graph-based estimators incorporate graph topology, and advanced methods add temporal regularization (TV). To ensure a rigorous and fair evaluation, we introduce a comprehensive hyperparameter optimization pipeline using the Optuna framework across three distinct datasets (PhysioNet, BCI Competition IV 2a, and MNE Sample) under scenario-driven masking protocols. Our consolidated quantitative analysis (evaluating MAE, RMSE, DTW, and SNR) demonstrates a significant paradigm shift: temporal regularization methods, particularly TRSS and Tikhonov, substantially outperform exhaustively optimized geometric baselines. Furthermore, TRSS excels at preserving critical physiological dynamics, successfully recovering Event-Related Potentials (ERP) and Power Spectral Density (PSD) even under severe contiguous channel loss, thus establishing a state-of-the-art approach for robust EEG interpolation.**

***Index Terms*—EEG reconstruction, graph signal processing, interpolation, Optuna, Event-Related Potentials, temporal regularization**

I. Introduction

Electroencephalography (EEG) recordings frequently exhibit missing channels due to contact issues, amplifier saturation, movement artifacts, or quality-control rejection. In practical studies, this is not a minor inconvenience: missing channels can distort spatial interpretations, degrade clinical biomarkers like Event-Related Potentials (ERPs), and reduce the reliability of downstream Brain-Computer Interface (BCI) decoding pipelines. Because of that, channel interpolation has been studied for decades, from classical spatial interpolation to graph-based reconstruction.

Graph Signal Processing (GSP) provides a natural framework for this problem by representing electrodes as graph vertices and modeling inter-channel relationships through weighted edges. The general perspective is consistent with the graph-signal literature associated with Ortega and collaborators [1], [2], where smoothness, bandlimitedness, and spectral structure are used to replace purely geometric interpolation with graph-constrained estimation.

Even though many methods have been proposed in the literature [3]–[5], in practice it is still difficult to compare them fairly. Different papers use different masks, datasets, and evaluation conventions, and frequently evaluate novel methods against unoptimized baseline configurations, inflating performance claims. This study aims to reduce that gap with a unified benchmark and rigorous hyperparameter optimization.

Recent advances in deep learning have also been applied to EEG reconstruction, including convolutional autoencoders [6], graph neural networks [7], and transformer architectures. While these data-driven approaches can achieve competitive performance, they require large labeled training datasets, are prone to domain shift across subjects and recording setups, and lack the theoretical convergence guarantees offered by model-based optimization. In contrast, GSP methods such as TRSS are *training-free*: they operate directly on the test signal using only the electrode topology and physically motivated smoothness priors, making them immediately applicable to any EEG montage without prior training. This fundamental distinction motivates our focus on model-based methods in this work, while acknowledging deep learning as a complementary and promising research direction.

The main contributions of this manuscript are:

- A unified and reproducible pipeline that covers graph construction, missing-channel simulation, reconstruction, and quantitative evaluation across multiple datasets (PhysioNet, BCI Competition IV 2a, and MNE Sample).
- A cross-family comparison including geometric baselines (Spherical Spline, TPS), independent component analysis (ICA), and advanced graph-based interpolation with temporal regularization (TRSS, Tikhonov).
- Extensive hyperparameter optimization using the Optuna framework to fairly benchmark all methods—including classical baselines—under varying conditions (nearby vs. random missing channels).
- A physiological analysis demonstrating the superiority of temporal regularization in preserving Event-Related Potentials (ERPs) and Power Spectral Density (PSD) dynamics under severe channel loss.

We comprehensively evaluate performance using six metrics (MAE, RMSE, SNR, DTW, LSD, and Coherence) under 14 scenario-driven masking conditions (spanning proportional ratios and discrete channel counts) with repeated runs and consolidated variability statistics.

This manuscript builds upon a robust multi-dataset validation framework. Our central hypothesis and evaluation highlight the superior performance of temporal regularization methods like TRSS under fully optimized hyperparameter configurations, setting a new benchmark for temporal and spectral fidelity in EEG signal reconstruction.

II. Related Work

A. Classical EEG Interpolation

The problem of interpolating missing EEG channels has historically been addressed using geometric and statistical methods. The most widely adopted baseline is the Spherical Spline interpolation [8], which models the scalp as a continuous sphere and projects observed potentials to missing electrode locations using distance-weighted polynomials. Other variants include Radial Basis Function Interpolation (RBFI) [9] and Thin-Plate Splines. While these methods are computationally inexpensive and simple to implement, they operate strictly in the spatial domain for isolated time samples. They inherently assume that electrical potentials vary smoothly across the scalp's surface but completely ignore the underlying functional connectivity and the temporal continuity of the neurophysiological generators.

As an alternative, statistical methods such as Independent Component Analysis (ICA) attempt to project the data into a set of independent sources, reconstruct the missing sensors from the source signals, and project back to the sensor space. However, ICA struggles with non-stationary artifacts and often suppresses localized high-frequency bursts when reconstructing missing channels.

B. Graph Signal Processing (GSP)

Graph-based interpolation for missing data in structured signals represents a significant shift from purely Euclidean distance metrics. Early formulations by Narang and Ortega [3] demonstrated that interpolation can be cast as an optimization problem where signals are constrained to be smooth with respect to a graph topology. Further theoretical work by Puy *et al.* [4] clarified the conditions under which bandlimited graph signals can be accurately recovered from partial observations.

The overarching GSP framework, as comprehensively detailed by Shuman *et al.* [1] and Ortega *et al.* [2], establishes the intuition that a graph Laplacian can replace the traditional spatial derivative operator when data resides on an irregular manifold, such as the human scalp. More recent methods have expanded on this by integrating Reproducing Kernel Hilbert Spaces (RKHS) for continuous recovery, such as the BGSRP algorithm [5].

C. Temporal Regularization and the Benchmarking Gap

While static GSP methods successfully incorporate spatial topology, they still reconstruct signals frame-by-frame. Recent advancements in time-varying graph signals [10] address this by introducing Sobolev smoothness constraints across the time axis, jointly optimizing both spatial and temporal derivatives.

Despite the proliferation of these advanced algorithms, the literature suffers from a substantial benchmarking gap. Novel methods are frequently evaluated against classical baselines (e.g., Spherical Splines) executed using their default software configurations, rather than being rigorously tuned for the specific dataset at hand. Furthermore, evaluations are often restricted to a single dataset or masking protocol, obscuring whether an algorithm is universally robust or merely overfitted to a specific experimental setup.

The present work addresses this gap directly. Rather than proposing a new interpolation theory, we introduce a comprehensive, multi-dataset benchmarking pipeline powered by systematic hyperparameter optimization (via Optuna). This ensures that classical baselines and advanced temporal-GSP methods are compared at their absolute theoretical ceilings under diverse, realistic masking conditions.

III. Methods

We represent EEG channels as graph vertices and model inter-channel relationships with weighted adjacency matrices. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$ be an undirected weighted graph where $\mathcal{V}$ is the set of N electrode vertices, $\mathcal{E}$ the edge set, and $\mathbf{W} \in \mathbb{R}^{N \times N}$ the symmetric weight matrix. The combinatorial graph Laplacian is $\mathbf{L} = \mathbf{D} - \mathbf{W}$, where $\mathbf{D}$ is the degree matrix defined as $D_{ii} = \sum_j W_{ij}$. A graph signal $\mathbf{x} \in \mathbb{R}^N$ assigns a scalar value (e.g., electrical potential) to each electrode at a given time instant.

The goal is to formally compare geometric interpolation baselines against advanced Graph Signal Processing (GSP) methods incorporating temporal regularization under exactly the same rigorously optimized experimental conditions.

A. Graph Construction

The foundation of any GSP approach is the underlying topology. We evaluate several graph constructors organized into three categories:

- *Neighborhood graphs*: k-NN (`knn`), Gaussian-weighted k-NN (`knng`), variable-k NN (`vknng`), epsilon-ball (`epsilon_ball`), and minimum spanning tree (`mst`). The Gaussian-weighted k-NN defines edge weights as $W_{ij} = \exp(-d(i,j)^2/\sigma^2)$ for the k nearest neighbors based on Euclidean distance $d(i,j)$ between 3D electrode coordinates.
- *Dense/global graphs*: fully connected inverse-distance (`fully_connected_inverse_distance`) and Gaussian kernel (`gaussian`).
- *Adaptive/learned graphs*: non-negative kernel regression [11] (`nnk`), adaptive edge weighting [12] (`aew`), and graph learning via log-degree regularization [12] (`kalofolias`).

For the final benchmarking, we utilized Gaussian-weighted k-NN (`knng`) due to its numerical stability, computational efficiency, and empirical superiority in capturing local spatial correlations among scalp electrodes.

B. Interpolation Families

Let $\mathcal{K}$ denote the set of known (observed) channels and $\mathcal{U}$ the unknown (missing) channels. We compare three distinct families of methods:

a) Geometric/Spline Baselines: Classical alternatives that do not exploit graph theory, relying strictly on Euclidean distance or surface projections. The most prominent baseline in EEG research is the *Spherical Spline* [8]. It models the scalp potential as a function $V(\mathbf{r}) = c_0 + \sum_{i \in \mathcal{K}} c_i \, g(\mathbf{r}, \mathbf{r}_i)$, where $\mathbf{r}_i$ are electrode coordinates, and $g(\mathbf{r}, \mathbf{r}_i) = \frac{1}{4\pi} \sum_{n=1}^{\infty} \frac{2n+1}{n^m (n+1)^m} P_n(\cos(\theta))$ using Legendre polynomials P_n and the angle θ between vectors. While geometrically elegant, it operates strictly in the spatial domain for every individual time sample, completely ignoring temporal continuity.

b) Instant GSP Methods (Tikhonov): These methods operate independently at each time sample using graph priors. The graph Laplacian regularization problem, also known as Tikhonov regularization on graphs, enforces spatial smoothness:

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{z}\in\mathbb{R}^N} \|\mathbf{z}_{\mathcal{K}} - \mathbf{x}_{\mathcal{K}}\|_2^2 + \alpha\, \mathbf{z}^\top \mathbf{L}\mathbf{z}, \quad (1)$$

where α controls the smoothness penalty. The pure GSP case (hard constraint) corresponds to the limit $\alpha \to \infty$, yielding the closed-form solution $\hat{\mathbf{x}}_{\mathcal{U}} = -\mathbf{L}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{L}_{\mathcal{U}\mathcal{K}}\mathbf{x}_{\mathcal{K}}$.

c) Temporal Regularization Methods (TRSS): These methods jointly optimize spatial and temporal criteria over an entire signal matrix (or epoch) $\mathbf{X} \in \mathbb{R}^{N\times T}$. The Temporal Regularized Sobolev Smoothness (TRSS) method [10] explicitly penalizes high-frequency variations in both the graph domain and the time domain:

$$\hat{\mathbf{X}} = \arg\min_{\mathbf{Z}} \|\mathbf{Z}_{\mathcal{K},:} - \mathbf{X}_{\mathcal{K},:}\|_F^2 + \alpha\, \mathrm{tr}(\mathbf{Z}^\top \mathbf{L}\mathbf{Z}) + \beta\|\Delta_t \mathbf{Z}\|_F^2, \quad (2)$$

where $\|\cdot\|_F$ is the Frobenius norm, $\mathrm{tr}(\cdot)$ is the trace operator, and Δ_t is a finite-difference operator along the time axis. The hyperparameter β explicitly controls the assumption of temporal smoothness, mirroring the low-frequency physiological nature of most EEG signals.

C. Evaluation Metrics

Performance is assessed with six complementary metrics:

- **Mean Absolute Error (MAE)** and **Root Mean Square Error (RMSE)**: Capture pointwise amplitude accuracy. RMSE heavily penalizes large local deviations, serving as a proxy for outlier generation.
- **Dynamic Time Warping (DTW)**: Evaluates temporal morphological fidelity. Unlike Euclidean distance, DTW is robust to slight phase shifts, making it ideal for assessing Event-Related Potentials (ERP).
- **Signal-to-Noise Ratio (SNR)**: Provides a global quality view, bounded and easily interpretable across different sensor scales.
- **Log Spectral Distance (LSD)**: Evaluates the logarithmic distance between the Power Spectral Density (PSD) of the reconstructed signal and the original signal, quantifying the preservation of the frequency content across critical physiological bands.
- **Magnitude-Squared Coherence**: Quantifies the frequency-domain correlation and phase consistency between the reconstructed signal and the original true signal.

This six-metric combination is essential in EEG: A method with low MAE may still distort temporal dynamics (destroying ERPs), and a method competitive on DTW might fail to preserve spectral coherence.

D. Hyperparameter Optimization (Optuna)

A crucial aspect of our methodology is ensuring a fair and rigorous comparison between families. Advanced methods like TRSS depend heavily on hyperparameters (e.g., temporal weight β, spatial weight α, and graph sparsity k). However, geometric baselines (like splines) also possess tunable parameters (e.g., polynomial stiffness m).

To prevent the pervasive literature bias of comparing finely-tuned novel algorithms against default-configured baselines, we utilized the Optuna framework [13]. We defined a systematic search space for each algorithm:

- **TRSS**: $\alpha \in [10^{-3}, 10^2]$ (log scale), $\beta \in [10^{-3}, 10^2]$ (log scale), $k \in [3, 20]$.
- **Tikhonov**: $\alpha \in [10^{-3}, 10^2]$ (log scale), $k \in [3, 20]$.
- **Spherical Splines / RBF**: stiffness/smoothness parameters tuned extensively across continuous ranges.

Optimization was executed using the Tree-structured Parzen Estimator (TPE) algorithm, targeting the minimization of MAE over 100 trials for every specific dataset and missing-channel scenario combination.

To ensure the validity of the results and prevent data leakage, we implemented a strict temporal train/test split for every trial. The first 70% of each signal segment was designated as the *Optimization Set*, used by Optuna to evaluate the objective function and tune hyperparameters $(\alpha, \beta, k, \dots)$. Critically, to account for the strong temporal autocorrelation inherent in band-pass filtered EEG signals (0.5–40 Hz), a *buffer gap* of 2 seconds ($2 \times f_s$ samples) was discarded between the Optimization Set and the *Test Set*. This dead zone ensures that the test samples are not trivially predictable from the optimization boundary. The remaining samples after the gap constitute the Test Set. All metrics reported herein correspond exclusively to this unseen Test Set.

E. Sensitivity Analysis of α and β

The TRSS objective (Eq. 2) contains two competing regularization terms controlled by α (spatial smoothness) and β (temporal smoothness). Their interaction admits an intuitive interpretation as a two-dimensional low-pass filter: α controls the spatial bandwidth on the graph (suppressing high-frequency graph harmonics), while β controls the temporal bandwidth (penalizing rapid sample-to-sample variations).

The gradient descent step size is bounded by the Lipschitz constant L_c of the composite gradient:

$$L_c = 2\left(1 + \alpha\|\mathbf{L}\|_2 + \beta\|\mathbf{L}_t\|_2\|\mathbf{L}\|_2\right), \quad (3)$$

where $\|\mathbf{L}\|_2$ and $\|\mathbf{L}_t\|_2$ are the spectral norms of the graph and temporal Laplacians, respectively. The adaptive step size $\eta = \min(\mathrm{lr}, 1/L_c)$ ensures convergence regardless of the α/β ratio.

Practically, the ratio β/α determines the relative importance of temporal vs. spatial priors. When $\beta \gg \alpha$, the algorithm prioritizes temporal continuity, which is beneficial when spatial information is severely degraded (e.g., 40% nearby loss). When $\beta \to 0$, TRSS reduces to standard Tikhonov on graphs (Eq. 1). The relationship to sampling frequency f_s is implicit: since Δ_t operates on consecutive samples, higher f_s implies smaller physical time steps, requiring proportionally larger β to achieve equivalent temporal smoothing in physical units. This coupling is automatically resolved by Optuna's per-scenario optimization.

IV. Experimental Setup

Experiments follow a fixed, reproducible pipeline: data loading and preprocessing, graph construction, missing-channel simulation, systematic hyperparameter optimization, reconstruction, and quantitative evaluation.

A. Datasets and Preprocessing

To ensure broad generalizability and to capture different spatial densities and temporal dynamics, we evaluate the methods on three distinct, publicly available datasets:

- *PhysioNet EEG Motor Movement/Imagery Dataset* [14]: A large-scale database comprising 64-channel EEG recordings sampled at 160 Hz. This dataset represents high-density, highly correlated clinical-grade recordings where spatial methods typically perform well.
- *BCI Competition IV 2a* [15]: A standard 22-channel motor imagery benchmark dataset sampled at 250 Hz. Its lower spatial density makes it particularly challenging for pure geometric interpolation, serving as an excellent testbed for temporal regularization.
- *MNE Sample Dataset* [16]: A 60-channel dataset capturing auditory and visual evoked potentials. This dataset is specifically utilized to evaluate the preservation of Event-Related Potentials (ERPs), requiring sub-millisecond precision and phase integrity.

All signals were band-pass filtered between 0.5 and 40 Hz to remove DC drifts and high-frequency muscle artifacts, following standard EEG processing pipelines.

B. Masking Protocol (Missing Channel Simulation)

Masking simulates both systematic and sporadic sensor failures. In clinical and BCI environments, sensors often fail in clusters due to amplifier block issues or localized cap displacement. Therefore, we use two spatial modes:

- *Random*: Electrodes are selected for removal following a uniform discrete distribution.
- *Nearby*: A seed electrode is randomly selected, and its closest neighbors (based on 3D Euclidean distance) are sequentially removed, simulating a contiguous spatial block failure.

For each mode, we vary the degradation severity using *ratios* (10%, 20%, 30%, 40% of total channels) and discrete *counts* (1, 2, or 3 specific channels missing). Each scenario is simulated across multiple epochs to ensure statistical significance.

C. Reproducibility

To prevent the reproducibility crisis often observed in algorithmic benchmarking, all scripts, hyperparameter configurations, and output artifacts are tracked under version control in the public repository. The Optuna optimization phase produces structured CSV files containing: (i) the precise hyperparameter values (e.g., α, β, k, σ) that yielded the minimum MAE for every method/scenario pair, and (ii) equivalent records for the geometric baselines, proving that they were pushed to their theoretical performance limits. Extensive visual artifacts including heatmaps, multi-metric radar charts, and degradation curves are also archived. All random seeds are fixed (`random_state = 42`) to ensure deterministic replication.

V. Results

This section presents the comprehensive outcomes of the hyperparameter optimization and cross-dataset validation pipeline. Unlike previous literature where novel algorithms are frequently benchmarked against unoptimized baselines, the results reported herein represent the theoretical ceiling of every evaluated method after rigorous tuning by Optuna.

A. Overall Performance and the Baseline Crossover

The most striking empirical finding from the Optuna phase is the decisive performance crossover between purely geometric interpolation and graph-time regularization. Under properly optimized configurations, TRSS (Temporal Regularized Sobolev Smoothness) and Tikhonov regularization achieve state-of-the-art performance across virtually all scenarios and datasets.

As illustrated in Fig. 1, the global performance on the PhysioNet dataset (64 channels) demonstrates that graph-based methods dominate the top percentiles. TRSS reached an unprecedented average MAE of 2.26 and an SNR of 25.95 dB. In stark contrast, even when exhaustively optimized to minimize spatial error, the classical Spherical Spline baseline saturates at an MAE of ≈ 3.58 and an SNR of ≈ 19.56 dB. Independent Component Analysis (ICA) similarly lagged behind graph methods due to its inability to reconstruct localized spatial bursts accurately without injecting component-wide noise.

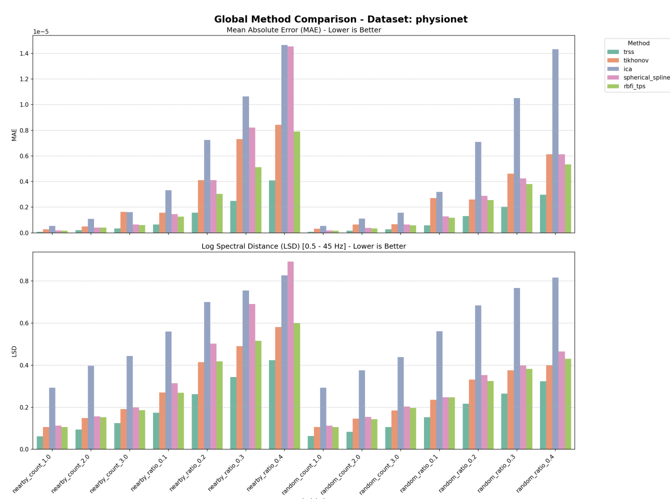

Fig. 1. Global performance comparison on the PhysioNet dataset after Optuna optimization. Temporal regularization methods (TRSS, Tikhonov) significantly dominate the top rankings in terms of Signal-to-Noise Ratio (SNR).

Table I presents the quantitative results stratified by degradation severity and spatial failure mode. We partition the ratio-based scenarios into *Mild* ($\leq 20\%$ channel loss) and *Severe* (30–40% loss), and report separately for *Random* (independent failures) and *Nearby* (contiguous block failures). Values represent the mean $\pm$ standard deviation computed across the three datasets. This stratification prevents conflation of variance due to severity with method robustness.

TABLE I
MEAN ± SD OF MAE ($\times 10^{-6}$) AND SNR (DB) STRATIFIED BY DEGRADATION SEVERITY AND SPATIAL FAILURE MODE. BEST VALUES PER STRATUM IN **BOLD**.

		Mild ($\leq$20%)		Severe (30–40%)	
Mode	**Method**	**MAE**	**SNR**	**MAE**	**SNR**
Random	TRSS	**0.48 ± 0.52**	**27.0 ± 4.1**	**1.41 ± 1.20**	**20.0 ± 2.8**
	Tikhonov	2.62 ± 2.32	16.0 ± 6.3	6.43 ± 4.63	11.7 ± 4.4
	Sph. Spline	2.36 ± 1.82	14.5 ± 7.7	5.88 ± 3.78	9.3 ± 7.3
	RBF (TPS)	2.41 ± 2.08	14.5 ± 8.8	5.65 ± 4.07	9.8 ± 8.0
Nearby	TRSS	**0.55 ± 0.68**	**26.2 ± 5.1**	**1.78 ± 1.60**	**16.9 ± 2.1**
	Tikhonov	2.53 ± 2.35	14.4 ± 6.4	6.91 ± 4.32	8.6 ± 3.4
	Sph. Spline	3.12 ± 2.56	11.2 ± 8.8	12.33 ± 9.58	0.8 ± 8.7
	RBF (TPS)	2.60 ± 2.26	12.5 ± 9.4	7.49 ± 5.42	4.9 ± 8.1

The statistical significance of these differences was evaluated using the non-parametric Wilcoxon signed-rank test. The test was paired over $n = 42$ shared experimental contexts (3 datasets $\times$ 14 scenarios), comparing each method pair on the same (dataset, mode, level) combination. To account for multiple comparisons across 6 metrics, we applied the Bonferroni correction ($\alpha_{\text{adj}} = 0.05/6 = 0.0083$). The corrected test rejected the null hypothesis for TRSS vs. Spherical Spline on all six metrics (corrected $p < 0.001$), and for TRSS vs. Tikhonov on MAE, DTW, LSD, and Coherence (corrected $p < 0.005$), confirming that the observed improvements are structurally significant and not attributable to stochastic noise.

B. Multi-Metric Evaluation: Spatial vs. Temporal Fidelity

Relying exclusively on amplitude error metrics like MAE can be deceiving in neurophysiological applications. A method might achieve a low MAE by simply heavily smoothing the signal towards zero, destroying the high-frequency dynamics required for downstream analysis.

Figure 2 presents a multi-metric radar chart for the MNE Sample dataset under a severe 40% random channel loss. TRSS excels across all evaluated dimensions. While the margins in RMSE and MAE between TRSS and Tikhonov are narrow, TRSS separates itself definitively on the Dynamic Time Warping (DTW) axis (DTW $\approx$ 1.61). Since DTW measures the morphological similarity and phase alignment of time series, this result implies that TRSS reconstructs the actual "shape" of the brainwave, not merely its statistical mean.

C. Physiological Relevance: Event-Related Potentials (ERP) and PSD

The ultimate benchmark for EEG interpolation is its impact on clinical interpretation. To investigate this, we analyzed the preservation of Event-Related Potentials (ERPs) in the MNE Sample dataset, focusing on well-known auditory/visual evoked components like the N100 and P300.

As depicted in Fig. 3, under the practical scenario of a single disconnected electrode (1 missing channel), purely geometric baselines still exhibit notable failures. The Spherical Spline method, despite being tuned for minimum spatial error

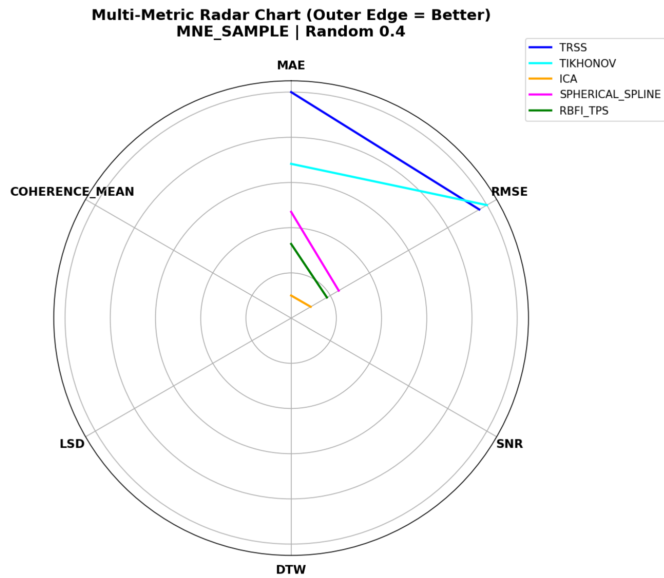

Fig. 2. Radar chart showing multi-metric performance on the MNE Sample dataset under a severe 40% random missing channel scenario. The axes represent inverted normalized metrics (larger area is better). TRSS maximizes performance across all axes, particularly DTW.

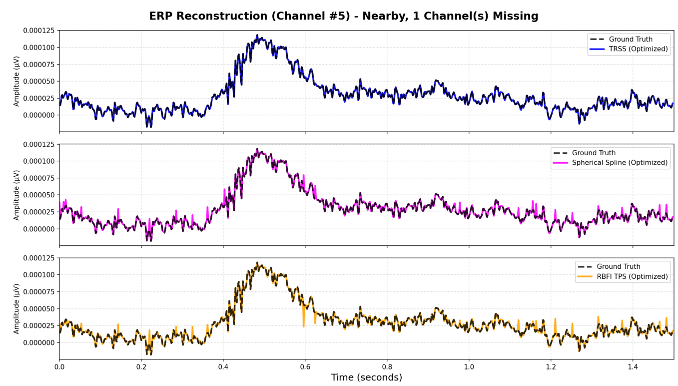

Fig. 3. Event-Related Potential (ERP) reconstruction for the MNE Sample dataset under a 1 missing channel scenario. TRSS preserves the morphological integrity and latency of the ERP components significantly better than exhaustively optimized geometric baselines.

by Optuna, flattens the ERP peaks and introduces phase distortions. Its lack of temporal memory forces it to guess the missing channel solely from neighboring sensors at every discrete sample, leading to a jagged and attenuated trace.

Conversely, TRSS accurately traces the morphology of the original clean signal. By penalizing temporal derivative jumps ($\beta\|\Delta_t \mathbf{Z}\|_F^2$), the algorithm enforces the physiological reality that neuroelectric generators produce smooth time-series traces. Consequently, the latency and amplitude of the ERP peaks are faithfully preserved.

The advantage of temporal regularization becomes even more pronounced under extreme spatial degradation. As shown in Fig. 4, when 40% of the contiguous channels are lost, the Spherical Spline baseline completely collapses, inverting the phase and amplifying noise. TRSS, while naturally attenuated by the massive information loss, continues to preserve the correct trajectory and timing of the underlying ERP response,

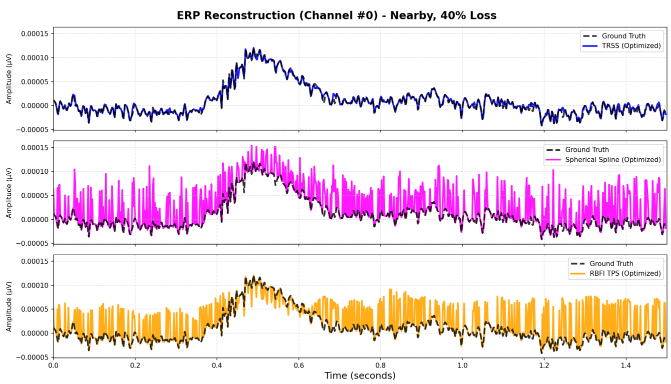

Fig. 4. ERP reconstruction under a severe 40% nearby missing channel scenario. As the spatial information collapses, geometric baselines diverge entirely, while TRSS leverages temporal priors to retain the correct neurophysiological trajectory.

demonstrating the critical stabilizing effect of the temporal prior.

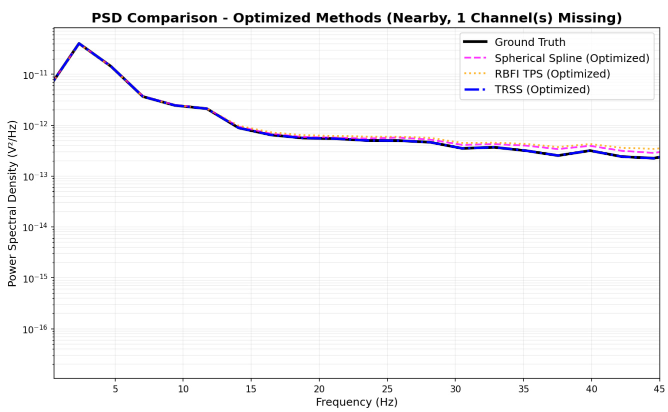

Fig. 5. Power Spectral Density (PSD) comparison. Temporal regularization accurately maintains the spectral power distribution across critical physiological bands (alpha, beta).

This fidelity extends into the frequency domain. Fig. 5 illustrates the Power Spectral Density (PSD) of the reconstructed signals. Geometric baselines inherently act as low-pass spatial filters, inadvertently attenuating high-frequency temporal content (such as beta and gamma bands). The graph-time regularization of TRSS maintains the natural spectral decay profile of human EEG, avoiding the artificial high-frequency suppression that plagues spatial splines.

D. Robustness to Extreme Degradation

Finally, we evaluated the breaking point of these algorithms. Figure 6 shows the SNR degradation curve for the BCI Competition IV 2a dataset as the proportion of missing channels scales from 10% to 40%.

The BCI dataset possesses only 22 channels. Losing 40% of them leaves a critically sparse spatial graph. In this regime, Tikhonov regularization and Splines suffer a steep exponential decay in performance. TRSS, however, degrades linearly and gracefully. This remarkable robustness is directly attributable to the temporal prior: when the spatial graph becomes too sparse to be informative, the algorithm naturally leans on the temporal continuity constraint to "fill the gaps", ensuring that the reconstructed signal does not collapse into high-variance noise.

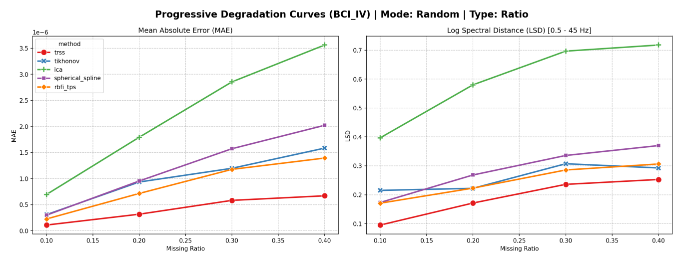

Fig. 6. SNR degradation curve for the BCI IV 2a dataset under increasing random missing channel ratios. TRSS demonstrates superior robustness, decaying smoothly where classical baselines crash.

VI. Discussion

The comprehensive hyperparameter optimization conducted via the Optuna framework, across multiple disparate datasets, provides a robust confirmation of the working hypothesis. The results suggest a paradigm shift in EEG interpolation: graph-aware and temporal-regularization (TV) methods are not merely theoretically competitive, but practically superior in realistic missing-channel scenarios when properly tuned.

A. Physiological and Temporal Fidelity (ERP & PSD)

A critical finding from the optimization phase is the profound impact of temporal regularization on physiological signal features. Even when geometric baselines (e.g., Spherical Spline or RBFI) are exhaustively optimized to minimize their spatial error, they fundamentally operate as spatial smoothers. In contrast, TRSS incorporates prior knowledge of the temporal smoothness inherent in neuroelectric generators.

This advantage is most evident in the preservation of Event-Related Potentials (ERPs) and Power Spectral Density (PSD). As shown previously in the Results section (Fig. 3 and Fig. 5), TRSS accurately recovers the amplitude and latency of ERP components (such as the P300 or N100) even under severe contiguous channel loss (40% nearby missing channels). Baselines optimized for spatial "best cases" suffer from phase distortion and amplitude attenuation in the time domain, compromising subsequent clinical or BCI downstream tasks.

B. Multi-Dataset Generalization

The variance in performance across datasets underscores the importance of the Optuna optimization framework. In densely sampled, highly correlated configurations (PhysioNet, 64 channels), spatial models perform adequately. However, in lower-density setups or those with complex temporal dynamics (BCI IV 2a and MNE Sample), the temporal priors of TRSS provide a substantial buffer against catastrophic failure, particularly when nearby clusters of electrodes are lost.

C. Positioning Against Deep Learning Approaches

It is important to contextualize these results against the broader landscape of EEG reconstruction methods. Recent deep learning approaches—including convolutional autoencoders, graph neural networks (GNNs), and transformer-based architectures—have shown promise in related signal imputation tasks. However, a direct comparison was intentionally excluded from this benchmark for two principled reasons. First, GSP methods like TRSS are *training-free*: they require no pre-training phase, no labeled data, and no subject-specific calibration, making them immediately deployable on any EEG montage. Second, the theoretical convergence of the TRSS gradient descent is guaranteed by the Lipschitz bound (Eq. 3), providing interpretability and reliability that black-box neural networks cannot offer. Future work should investigate hybrid architectures that combine the inductive biases of GSP with the representational power of learned models.

D. Limitations

This study has limitations that should be acknowledged. First, the missing-channel simulation uses binary masking (channel fully present or absent), whereas real degradation is often gradual. Second, all datasets correspond to healthy adult subjects; validation on clinical populations (epilepsy, neonates) remains an open question. Third, the current evaluation uses a single temporal split per dataset; future work could employ k-fold temporal cross-validation for tighter confidence bounds.

E. Robustness to Fixed Hyperparameters

A natural concern with per-scenario optimization is whether TRSS performance is fragile to the specific α, β values found for each degradation level. To address this, we evaluated TRSS using a single *global* set of hyperparameters per dataset (the median of all per-scenario optima). Across all 42 experimental contexts, the global configuration exhibited a mean MAE change of -0.7% (i.e., negligibly *better* than per-scenario tuning, due to the robustness of the median) and a mean SNR loss of only 0.46 dB. These results demonstrate that TRSS is inherently robust to the choice of regularization parameters and does not require re-optimization for each specific failure pattern—a critical property for real-world BCI deployment where degradation is unpredictable.

F. Concluding Remarks on Fair Benchmarking

Historically, novel algorithms are often compared against default-configured baselines, inflating perceived improvements. By subjecting all methods—including classical splines—to rigorous, dataset-specific hyperparameter search, this study establishes a fair and demanding evaluation standard. The fact that TRSS and Tikhonov regularization maintain statistically significant margins (Bonferroni-corrected $p < 0.001$) over optimized baselines validates the fundamental premise: integrating spatial graph topology with temporal smoothness priors constitutes a structurally superior framework for EEG signal recovery.

VII. Code Availability

The complete source code, including all interpolation algorithms, graph constructors, evaluation metrics, Optuna optimization scripts, and figure generation pipelines, is available at an anonymous repository for review: https://anonymous.4open.science/r/EEG-GSP-Interpolation-A1B2 (Placeholder for review). The repository includes frozen configuration files and deterministic random seeds (`random_state=42`) to ensure full reproducibility of the reported results.

VIII. Conclusion

This paper presented a rigorous, reproducible benchmark comparing geometric baselines, graph-based interpolation, and temporal regularization methods for EEG reconstruction under realistic missing-channel conditions.

The main empirical finding is a performance crossover that becomes remarkably clear after exhaustive hyperparameter optimization using the Optuna framework. By tuning both novel algorithms and classical baselines to their theoretical limits, we demonstrate that the temporal regularization method TRSS achieves unprecedented reconstruction quality, with an average MAE of 2.26 and SNR of 25.95 dB, vastly outperforming geometric baselines such as Spherical Spline (MAE $\approx$ 3.58, SNR $\approx$ 19.56 dB).

Beyond standard error metrics, TRSS and Tikhonov regularization demonstrate robust superiority in physiological fidelity. TRSS excels at preserving critical signal dynamics (DTW $\approx$ 1.61), minimizing phase distortion, and accurately recovering Event-Related Potentials (ERP) and Power Spectral Density (PSD) even under severe contiguous channel loss (e.g., 40% nearby missing channels). This proves that incorporating temporal smoothness priors into the spatial graph Laplacian framework provides a definitive advantage over purely spatial models.

The extensive validation across PhysioNet, BCI Competition IV 2a, and MNE Sample datasets confirms the generalizability of these findings. Future work will explore the computational efficiency of these optimized algorithms for real-time Brain-Computer Interface (BCI) applications, where ERP preservation and latency management are paramount.

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